

MISFIT: A Self-Supervised Pretraining Toolkit for 3D Medical Imaging

A Preliminary Test Across Three BraTS 2026 Segmentation Tasks

Adrian Celaya^{1,2}, Yogesh Kumar¹, Awj Twam¹, Beatrice Riviere², David Fuentes¹, Tucker Netherton¹

¹The University of Texas MD Anderson Cancer Center, Houston TX ²Rice University, Houston TX

Introduction

Despite continued architectural innovation in medical image segmentation, reported performance gains are not always reproducible, and standardized tooling for consistent comparisons remains scarce. We address this with **MIST**, our modular, end-to-end framework for training, testing, and evaluating segmentation methods in a standardized, reproducible manner — which placed **3rd** in the 2024 and 2025 BraTS challenges. For BraTS 2026 we enter three tasks — brain metastases, pediatric tumors, and cross-population generalizability — each testing a different kind of generalization.

This year we also debut **MISFIT**, our in-development framework for self-supervised pretraining of 3D medical-imaging encoders via masked autoencoding (MAE). We use it to pretrain a SwinUNETR-V2 encoder on a pooled, cross-task corpus of over 16,000 unlabeled MRI volumes spanning all three tasks' training data.

This report asks two questions

- 01

System-level: does a **MISFIT**-pretrained SwinUNETR-V2 pipeline beat our established pocket nnU-Net baseline?
- 02

Isolated effect: holding architecture, patch size, and budget fixed, does MAE pretraining itself help over random initialization?

Methods

Data

Table 1 · Dataset sizes and regions evaluated per task

TASK	COHORT	TRAIN	HELD-OUT	REGIONS
1	Brain metastases	1,196	179	NETC, SNFH, ET, RC
2	Pediatric tumors	294	91	ET, NET, CC, ED
3	BraTS-GoAT	1,351	450	ET, TC, WT

Self-supervised pretraining via MISFIT

MISFIT pretrains 3D medical-imaging encoders with masked autoencoding (MAE), independent of MIST. We pooled every training-split MRI volume across all three tasks into **16,315** single-channel volumes, masked 75% of each crop, and trained a SwinUNETR-base encoder for 200 epochs to reconstruct it. That encoder was then transferred into MIST's SwinUNETR-base segmentation architecture.

Three network arms

- Pocket nnU-Net (baseline)** — our returning conv baseline, 128³ patch, random init.
- SwinUNETR-V2, MISFIT-pretrained** — 96³ patch, encoder from MISFIT weights.
- SwinUNETR-V2, from scratch** — identical, but randomly initialized — the matched control.

The two SwinUNETR arms are matched on everything but initialization, isolating the pretraining effect. The baseline differs in architecture, patch size, *and* budget too, so it tests the full pipeline instead.

Training & evaluation

5-fold cross-validation; the baseline trained up to 5,000 epochs/fold (mean 3,107–4,252 actual), both SwinUNETR arms exactly 1,000. Arms are compared via BraTS-style average rank (MIST's `mist_rank`) and a two-sided Wilcoxon signed-rank test ($\alpha = 0.05$).

Discussion & Conclusion

Main finding — MAE pretraining helps once architecture, patch size, and budget are held fixed, but held-out validation does not support the pooled MISFIT pipeline beating our established baseline.

- Task 1's reversal is explainable.** Its lesion-wise scoring amplifies a small boundary deficit that per-voxel CV Dice hides; 1,000 epochs likely wasn't enough to refine it.
- Pretraining helps most where labels are scarce.** Task 2 (294 cases) sees the largest gain (ET Dice 0.518 → 0.670) and is the only task where the from-scratch model fails to converge.
- Final submission: pocket nnU-Net for all three tasks** — required by data-usage rules, and independently supported by the held-out results.
- This is a preliminary MAE schedule** — 200 epochs against the 800–1,600 convention. Longer pretraining may close the gap.

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Results

Table 2 · BraTS-style average rank (lower is better), two-sided Wilcoxon p-values

TASK	5-fold cross-validation			Held-out validation		
	BASE	MISFIT	P	BASE	MISFIT	P
1	1.503	1.497	0.62	1.427	1.573	1.5×10⁻⁴
2	1.528	1.472	0.026	1.503	1.497	0.83
3	1.518	1.482	0.0072	1.513	1.487	0.32
Task	MISFIT	Scratch	p	MISFIT	Scratch	p
1	1.439	1.561	2.1×10⁻⁵	1.436	1.564	0.0024
2	1.457	1.543	0.035	1.440	1.560	0.0053
3	1.416	1.584	1.9×10⁻¹³	1.424	1.576	7.4×10⁻⁵

Against the **matched control**, MISFIT pretraining wins on all three tasks under both protocols — the isolated effect is real. Against the **established baseline**, the picture reverses: CV favors the pretrained pipeline on Tasks 2–3, but that vanishes on held-out validation, where the baseline pulls significantly ahead on Task 1 ($p = 1.5\times 10^{-4}$).

Why CV doesn't replicate — the MAE encoder saw pooled training-split images label-free, inflating CV scores for Tasks 2–3. Held-out scoring excludes the pretraining pool entirely, which is why we treat it as decisive.

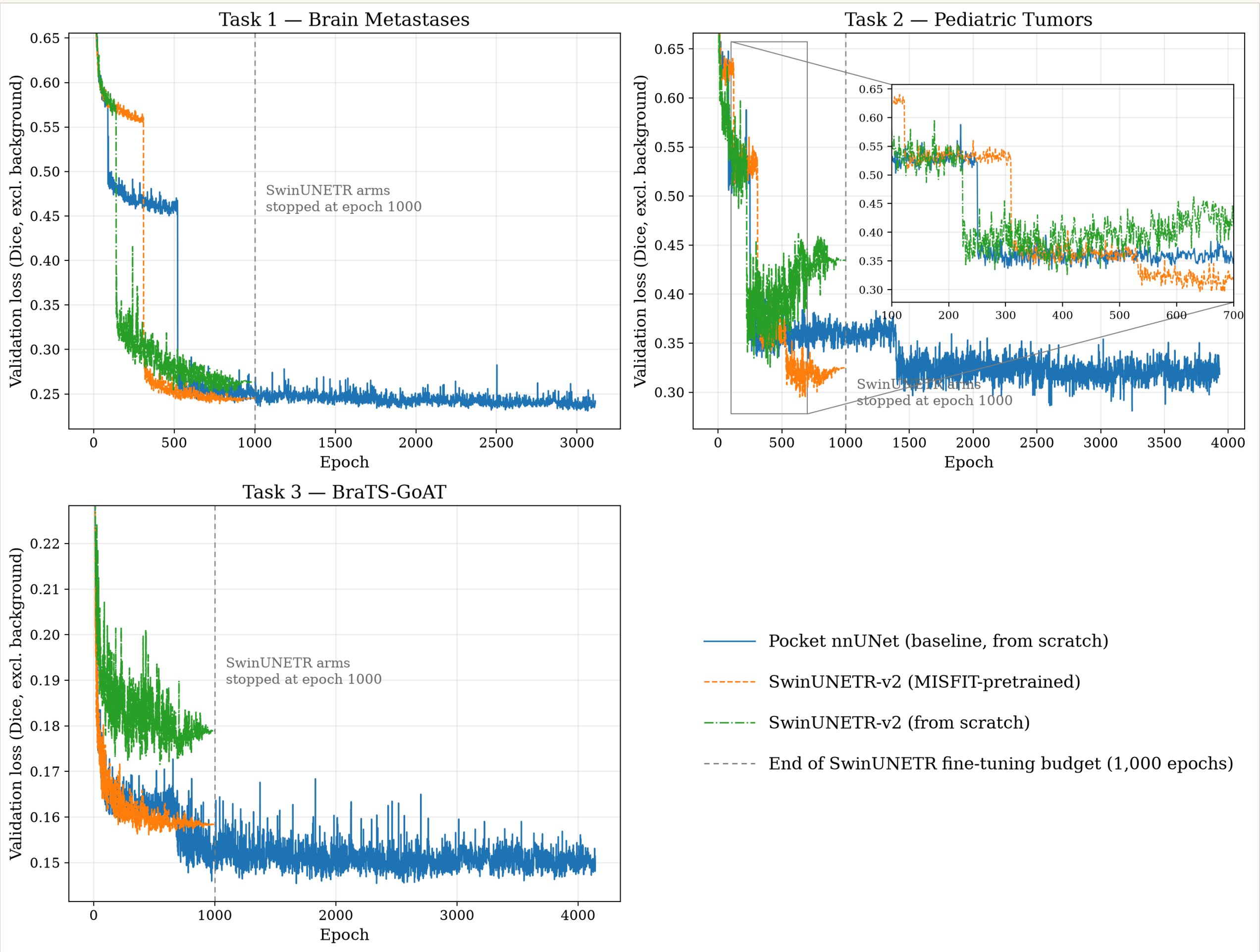


Figure 1. Validation loss (Dice, excl. background) across training, one representative fold per task. The MISFIT-pretrained arm reaches within 3.5–7.3% of the baseline's fully converged floor inside its 1,000-epoch budget, in far fewer epochs than the baseline needs (means: 3,107 / 3,908 / 4,252 epochs for Tasks 1–3). The from-scratch SwinUNETR converges but plateaus above both other arms on Tasks 1 and 3, and fails to converge at all on Task 2.

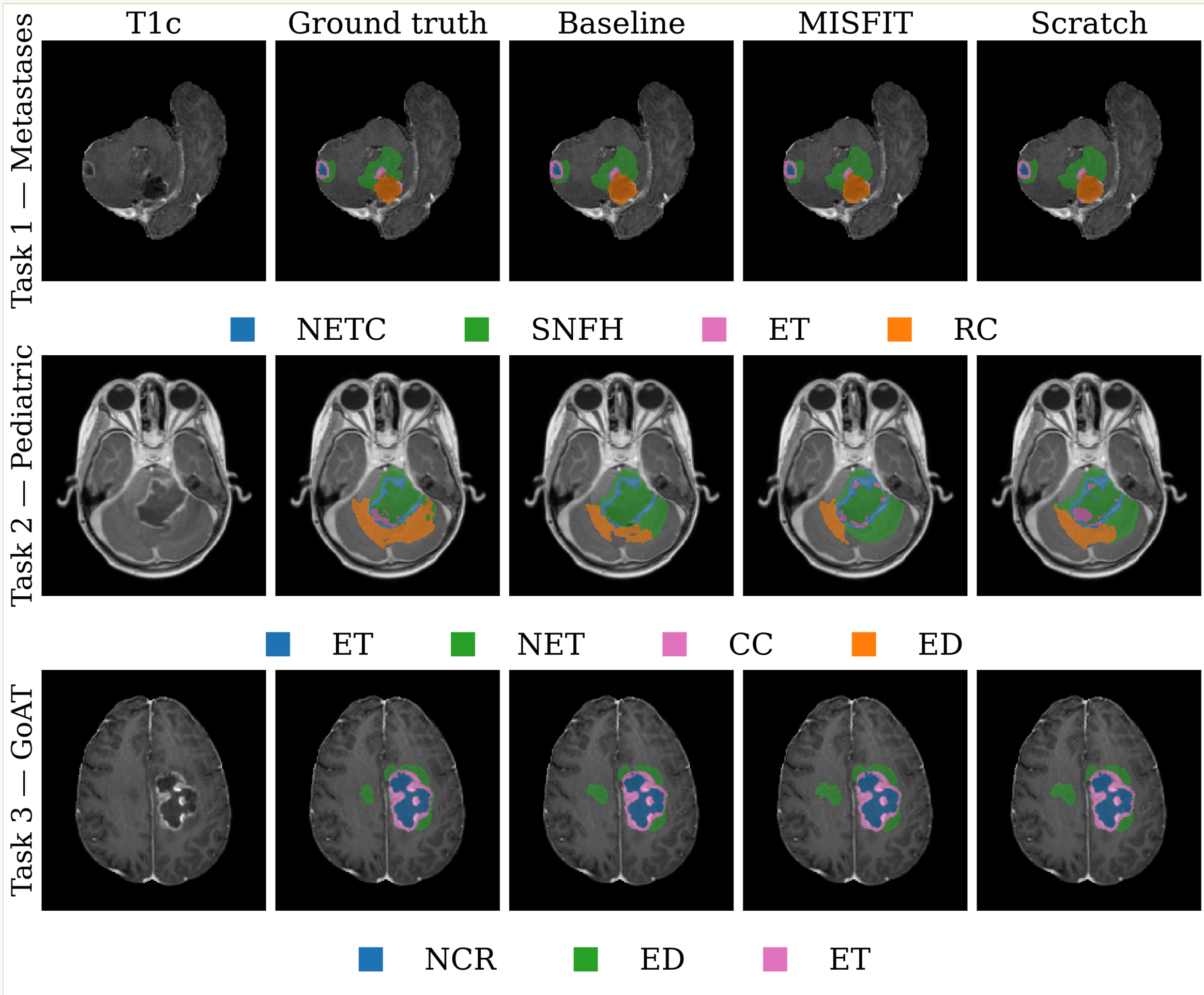


Figure 2. One representative training-cohort case per task, comparing T1c and ground truth against all three arms (baseline, MISFIT-pretrained, from-scratch). All three agree closely on Tasks 1 and 3; Task 2 is the only row where they visibly diverge — the baseline recovers almost none of the cystic component (pink).